

Potential Flow over Wings

Lift distribution

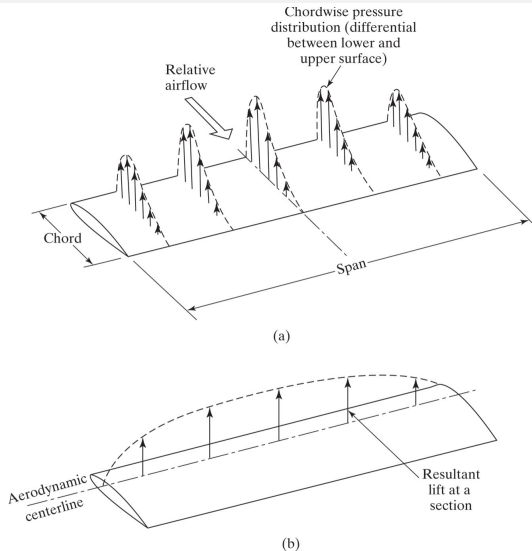


Figure 5.1: Flow over wings - qualitative effects, Fig. 7.1 from AE-BC

Wingtip vortices

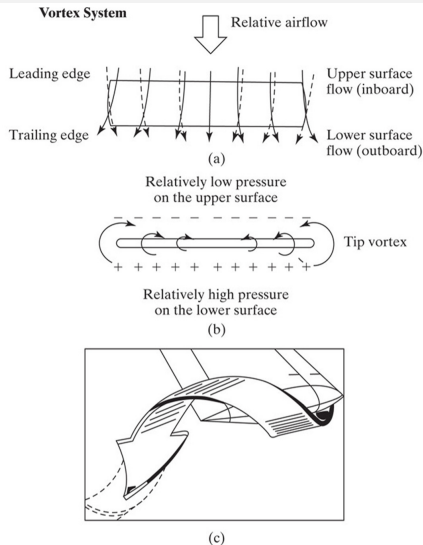
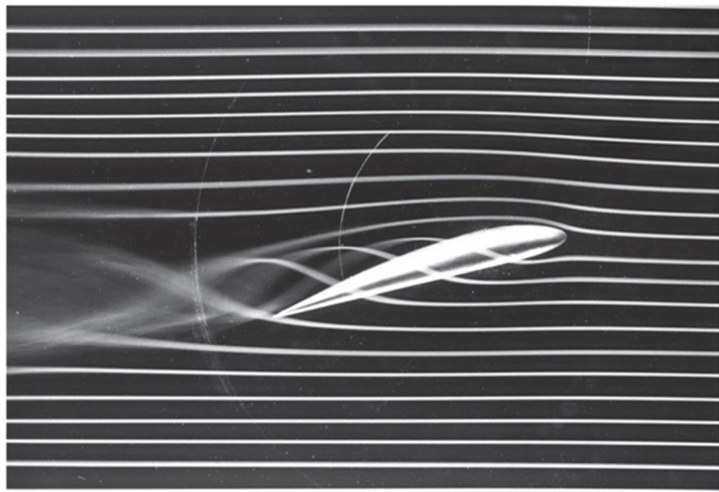


Figure 5.2: Flow over wings - qualitative effects, Fig. 7.2 from AE-BC

Wingtip vortices



(d)

Figure 5.3: Flow over wings - qualitative effects, Fig. 7.2 from AE-BC

Wingtip vortices



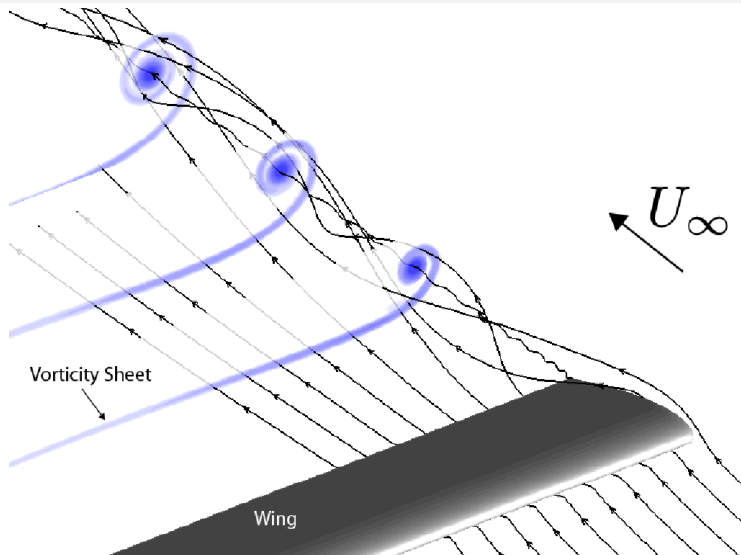
http://www.flytime.ca/files/2013/05/wingtip_vortices_flyingindiancom.jpg

Wingtip vortices



https://howthingsfly.si.edu/sites/default/files/image-large/il_wingtipvortexedit_lg.jpg

Wingtip vortices



https://upload.wikimedia.org/wikipedia/commons/1/1e/Tip_vortex_rollup.png

Downwash & Induced Drag

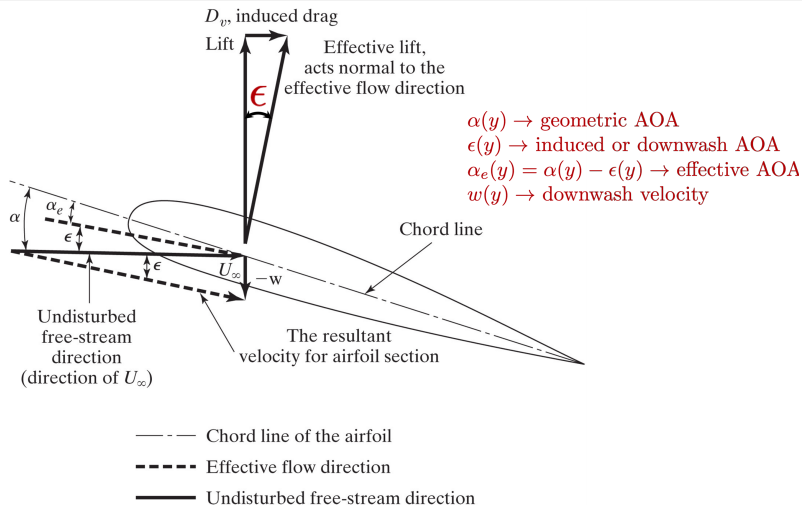


Figure 5.4: Downwash velocity and its effects, modified Fig. 7.6 from AE-BC

Comparison aspect ratio

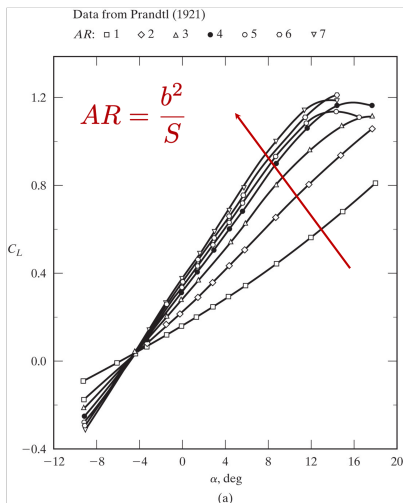


Figure 5.5: C_L vs. α aspect ratio effects, modified Fig. 7.9b from AE-BC

Note- These are from experiments and C_L is for a wing

Comparison aspect ratio

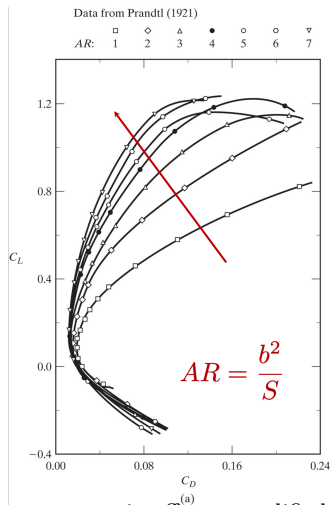


Figure 5.6: Drag polar aspect ratio effects, modified Fig. 7.9a from AE-BC

Note- These are from experiments and C_L is for a wing

Induced drag

We approximate the drag on a wing by

$$C_D = \underbrace{C_{D2-D}}_{\text{form+skin friction drag}} + \underbrace{C_{Dv}}_{\text{induced drag}} \quad (5.1)$$

Note: for potential flows $C_{D2-D} = 0$

Biot-Savart Law

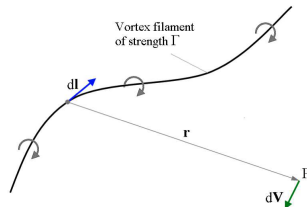


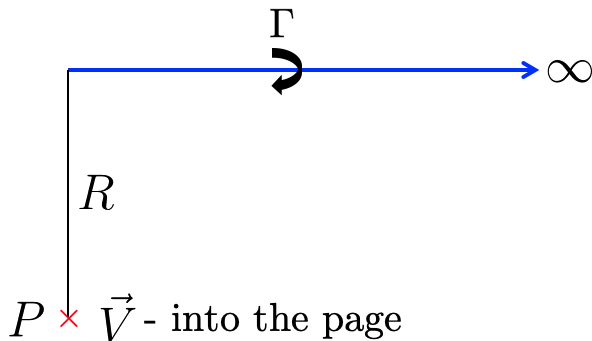
image adapted from Wikipedia

$$d\vec{V} = \frac{\Gamma}{4\pi} \frac{d\vec{l} \times \vec{r}}{|\vec{r}|^3} \quad (5.2)$$

$$\text{Induced velocity} \rightarrow \vec{V} = \frac{\Gamma}{4\pi} \int_C \frac{d\vec{l} \times \vec{r}}{|\vec{r}|^3} \quad (5.3)$$

Velocity induced by vortex line, hence induced velocity.

Biot-Savart Law



based on image from FA-JA

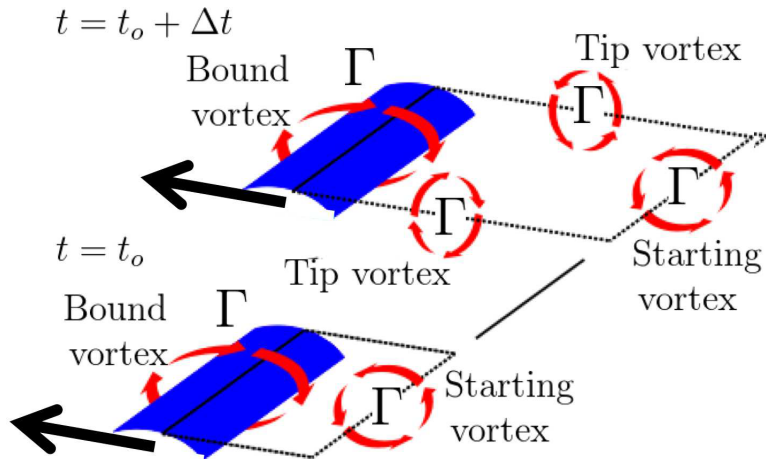
Velocity induced by vortex line extending to ∞ :

$$|\vec{V}| = \frac{\Gamma}{4\pi R} \quad (5.4)$$

Helmholtz' vorticity theorem

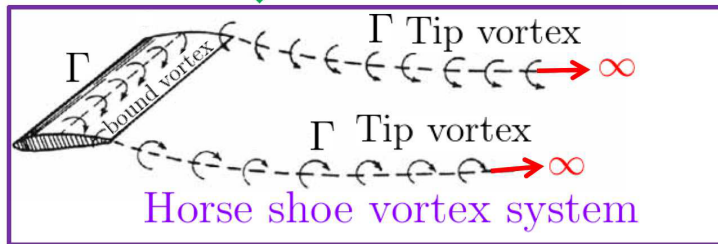
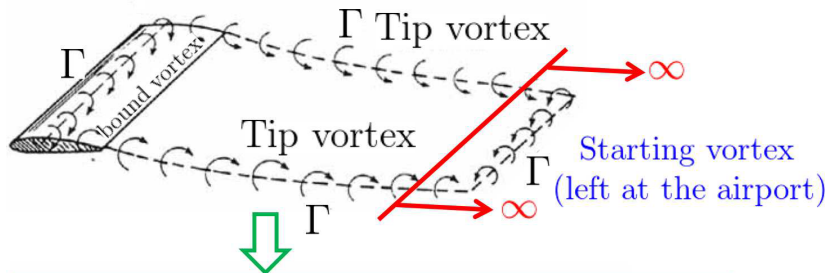
1. Circulation Γ is constant along a vortex tube or line
2. A vortex tube or line cannot end in the fluid. It must
 - (a) go to ∞
 - (b) terminate on a boundary
 - (c) close on itself

Starting & bound vortex



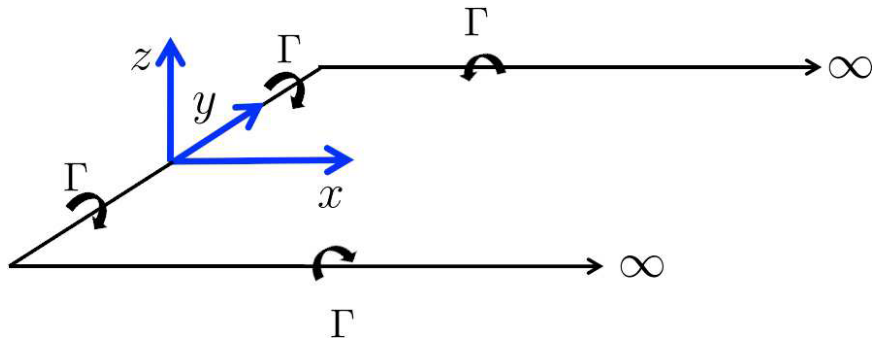
partially adapted from <http://www.onemetre.net/Design/Downwash/LiftLine/Line2.gif>

Horse shoe vortex model of wing

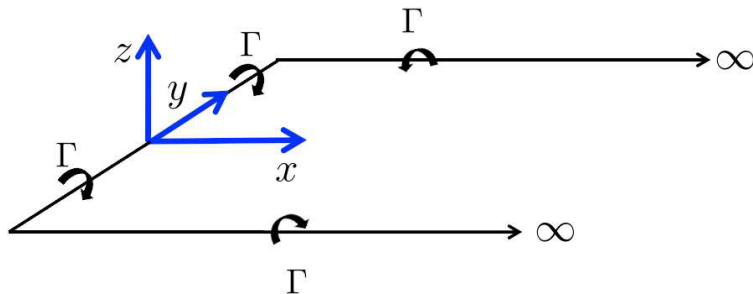


partially adapted from <http://history.nasa.gov/SP-367/f53.htm>

Horse shoe vortex model of wing



Problems with horse vortex model

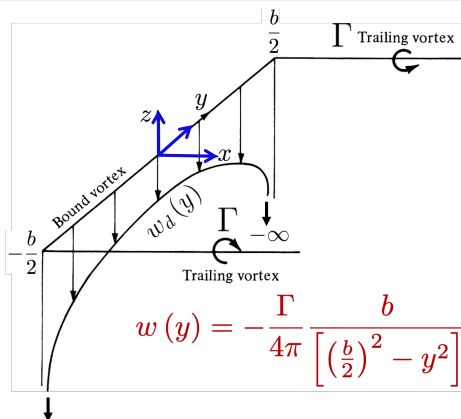


$$\text{single semi-infinite vortex} \rightarrow |\vec{V}| = \frac{\Gamma}{4\pi R} \quad (5.5)$$

Therefore, downwash distribution $w_d(y)$ for horse shoe system:

$$w(y) = -\frac{\Gamma}{4\pi} \frac{b}{\left[(b/2)^2 - y^2\right]} \quad (5.6)$$

Problems with horse vortex model



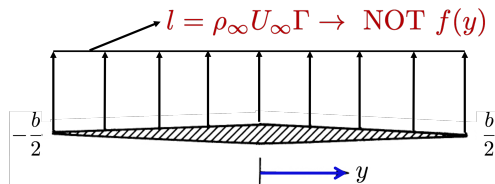
$$w(y) = -\frac{\Gamma}{4\pi} \frac{b}{\left[\left(\frac{b}{2}\right)^2 - y^2\right]}$$

partially adapted from FA-JA ∞

$$w\left(\pm \frac{b}{2}\right) = -\infty, \quad w(0) = -\frac{\Gamma}{\pi b} \quad \text{Horse shoe system - Not realistic} \quad (5.7)$$

$$(5.8)$$

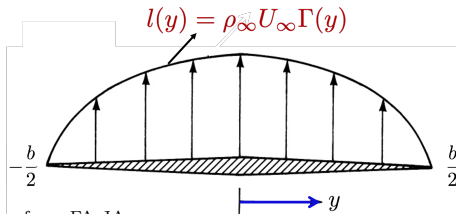
Problems with horse vortex model



$$l = \rho_\infty U_\infty \Gamma \rightarrow \text{not a function of } y \text{ (uniform loading)} \quad (5.9)$$

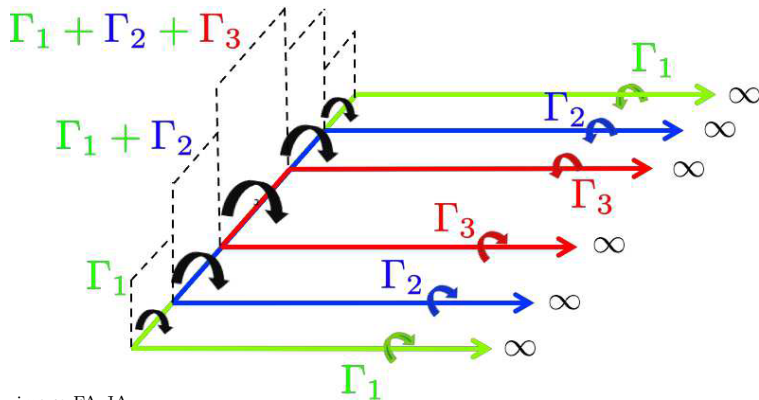
$$L = \rho_\infty U_\infty \Gamma b \rightarrow \text{not realistic} \quad (5.10)$$

$$l'(y) = \rho_\infty U_\infty \Gamma(y) \rightarrow \text{more realistic (non-uniform loading)} \quad (5.11)$$



both figures based on those from FA-JA

Prandtl's lifting line theory



based on image FA-JA

We can superpose multiple horse-shoe vortex systems to get a more realistic representation. We can use superposition because we are considering **potential flows**.

Prandtl's lifting line theory

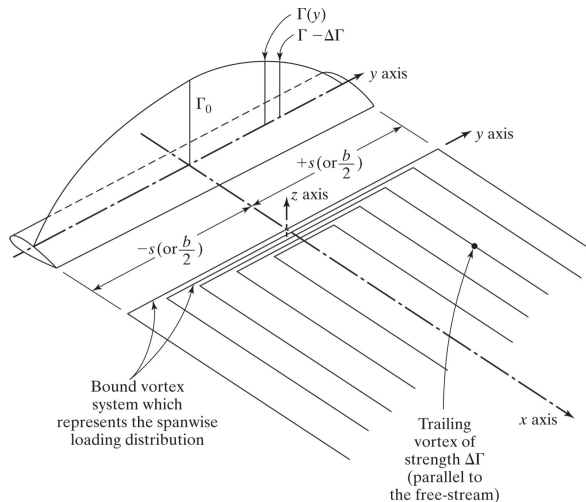


Figure 5.7: PLLT vortex system, Fig. 7.4a from AE-BC

Prandtl's lifting line theory



Figure 5.8: Vortex shedding behind a B747, Fig. 7.4b from AE-BC

Prandtl's lifting line theory

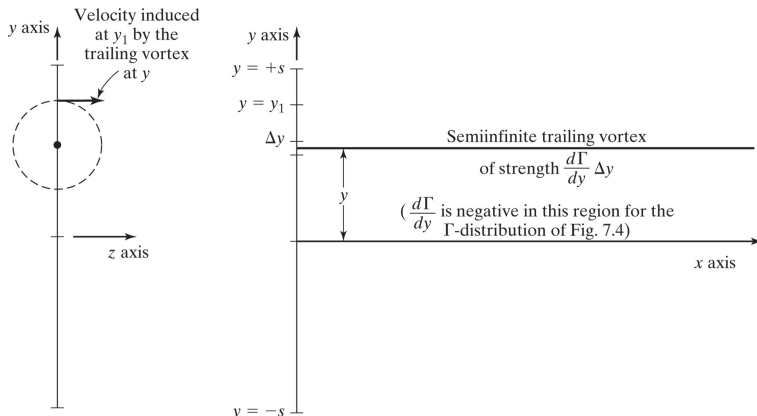


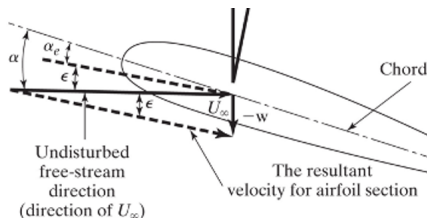
Figure 5.9: Induced velocity calculation, Fig. 7.5 from AE-BC
Downwash at y_1 due to vortex contribution from y

$$\delta w_{y1} = \frac{1}{2} \left[\frac{d\Gamma}{dy} dy \frac{1}{2\pi(y - y_1)} \right] \quad (5.12)$$

Prandtl's lifting line theory

Therefore, the downwash w at y_1 due to the entire vortex sheet is,

$$w(y_1) = \int_{-b/2}^{b/2} dw(y_0) = \frac{1}{4\pi} \int_{-b/2}^{b/2} \left(\frac{d\Gamma}{dy} \right) \frac{1}{y - y_1} dy \quad (\text{TEq. (7.2)}) \quad (5.13)$$

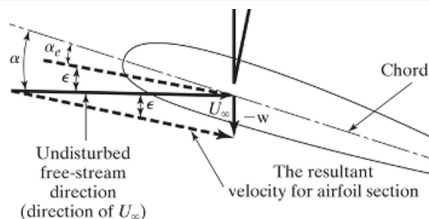


Using the figure, the induced or downwash AOA ϵ and the effective AOA α_e are

$$\tan \epsilon = \frac{-w}{U_\infty}, \text{ if } \epsilon \text{ is small } \tan \epsilon \approx \epsilon \approx -\frac{w}{U_\infty} \quad (\text{TEq. (7.3)}) \quad (5.14)$$

$$\alpha_e = \alpha - \epsilon \quad (\text{TEq. (7.4)}) \quad (5.15)$$

Prandtl's lifting line theory



Therefore substituting eq.(5.13) the induced AOA is

$$\epsilon = \frac{-w(y_0)}{U_\infty} = -\frac{1}{4\pi U_\infty} \int_{-b/2}^{b/2} \left(\frac{d\Gamma}{dy} \right) \frac{dy}{(y - y_1)} \quad (5.16)$$

$$\alpha_e = \alpha - \epsilon = \alpha + \frac{1}{4\pi U_\infty} \int_{-b/2}^{b/2} \left(\frac{d\Gamma}{dy} \right) \frac{dy}{(y - y_1)} \quad (5.17)$$

$$\alpha = \alpha_e - \frac{1}{4\pi U_\infty} \int_{-b/2}^{b/2} \left(\frac{d\Gamma}{dy} \right) \frac{dy}{(y - y_1)} \quad (5.18)$$

Prandtl's lifting line theory

From TAT (cambered airfoil:)

$$C_l(y_1) = 2\pi[\alpha_e(y_1) - \alpha_{ol}(y_1)] \quad (5.19)$$

$$l = \rho_\infty U_\infty \Gamma(y_1) = \frac{1}{2} C_l(y_1) U_\infty^2 c(y_1) \rho_\infty \quad (5.20)$$

$$C_l(y_1) = \frac{2\Gamma(y_1)}{U_\infty c(y_1)} \quad (5.21)$$

Sub eq. 5.21 in 5.19

$$\frac{2\Gamma(y_1)}{U_\infty c(y_1)} = 2\pi[\alpha_e(y_1) - \alpha_{ol}(y_1)] \quad (5.22)$$

$$\alpha_e(y_1) = \frac{\Gamma(y_1)}{\pi U_\infty c(y_1)} + \alpha_{ol}(y_1) \quad (5.23)$$

Prandtl's lifting line theory

Sub eq. 5.23 in 5.18

$$\alpha(y_1) = \underbrace{\frac{\Gamma(y_1)}{\pi U_\infty c(y_1)} + \alpha_{0l}(y_1)}_{\alpha_e} - \underbrace{\frac{1}{4\pi U_\infty} \int_{-b/2}^{b/2} \left[\frac{d\Gamma}{dy} \right] \frac{dy}{(y - y_1)}}_{\epsilon} \quad (5.24)$$

Fundamental Eqn. of Prandtl's lifting line theory:

$$\alpha(y_1) = \frac{\Gamma(y_1)}{\pi U_\infty c(y_1)} + \alpha_{0l}(y_1) + \underbrace{\frac{1}{4\pi U_\infty} \int_{-b/2}^{b/2} \left[\frac{d\Gamma}{dy} \right] \frac{dy}{(y_1 - y)}}_{\epsilon} \quad (5.25)$$

Solve for $\Gamma(y)$, Note y is the same as y_1 i.e., y_1 is a dummy variable

Prandtl's lifting line theory

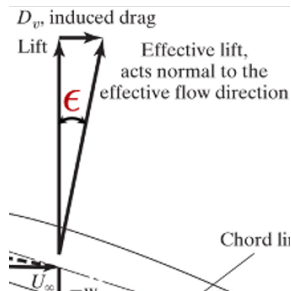
Once $\Gamma(y)$ is known:

$$1. \quad l(y) = \rho_{\infty} U_{\infty} \Gamma(y) \rightarrow \text{Load distribution} \quad (5.26)$$

$$2. \quad L = \int_{-b/2}^{b/2} l(y) \, dy = \int_{-b/2}^{b/2} \rho_{\infty} U_{\infty} \Gamma(y) \, dy \quad (\text{TEq. (7.6)}) \quad (5.27)$$

$$C_L = \frac{L}{q_{\infty} S} = \frac{2}{U_{\infty} S} \int_{-b/2}^{b/2} \Gamma(y) \, dy \quad (5.28)$$

Prandtl's lifting line theory



$$3. \quad d_v \approx l\epsilon \quad (\text{for small } \epsilon) \rightarrow d_v(y) \approx l(y)\epsilon(y)$$

$$= \rho_\infty U_\infty \Gamma(y) \left(\frac{-w(y)}{U_\infty} \right) = -\rho_\infty \Gamma(y) w(y) \quad (5.29)$$

$$D_v = \int_{-b/2}^{b/2} d_v \, dy = - \int_{-b/2}^{b/2} \rho_\infty \Gamma(y) w(y) \, dy \quad (\text{TEq. (7.7)}) \quad (5.30)$$

$$C_{Dv} = \frac{D_v}{q_\infty S} = \frac{2}{U_\infty^2 S} \int_{-b/2}^{b/2} \Gamma(y) w(y) \, dy \quad (5.31)$$

Elliptical lift distribution

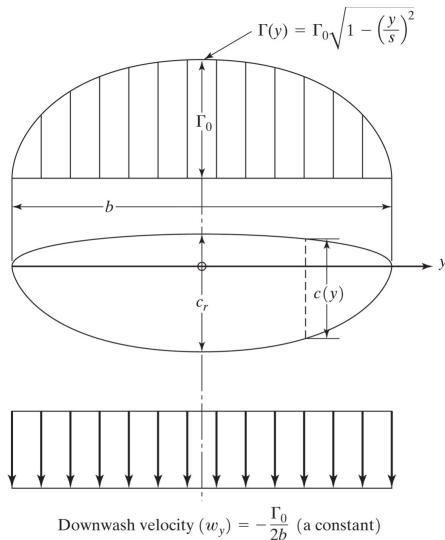


Figure 5.10: Elliptical lift distribution, Fig. 7.5 from AE-BC

Elliptical lift distribution

Elliptical load distribution:

$$\Gamma(y) = \Gamma_0 \sqrt{1 - \left(\frac{2y}{b}\right)^2} \quad (5.32)$$

$$y = 0 \text{ @ root, } \Gamma(y=0) = \Gamma_0, \quad y = \pm \frac{b}{2} = \pm s \text{ @ span } \Gamma\left(y = \pm \frac{b}{2} = \pm s\right) = 0$$

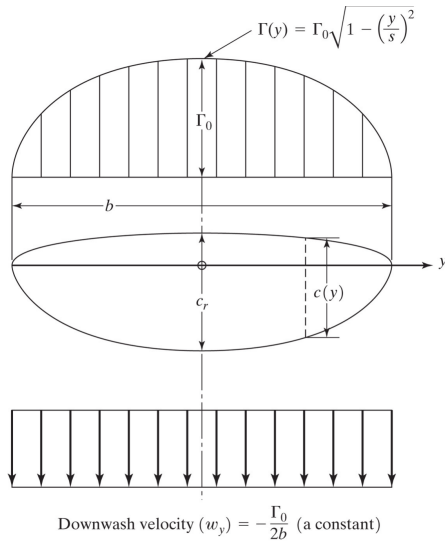
$$w(y_1) = \frac{1}{4\pi} \int_{-b/2}^{b/2} \left(\frac{d\Gamma}{dy}\right) \frac{1}{4\pi(y-y_1)} dy \quad (\text{TEq. (7.2)})$$

After “some” algebra

$$w_d(y_0) = -\frac{\Gamma_0}{2b} = -\frac{\Gamma_0}{4s} \quad (\text{TEq. (7.10)}) \quad (5.33)$$

Therefore, the downwash distribution for an elliptically loaded wing is constant

Elliptical lift distribution



Elliptical lift distribution

$$\epsilon = -\frac{w_d}{U_\infty} = \frac{\Gamma_0}{2bU_\infty} = \frac{\Gamma_0}{4sU_\infty} \quad (5.34)$$

Therefore, since the downwash distribution for an elliptically loaded wing is constant the induced angle of attack is also a constant

$$C_L = \frac{L}{q_\infty S} = \frac{2}{U_\infty S} \int_{-b/2}^{b/2} \Gamma(y) dy \quad (5.35)$$

$$= \frac{2}{U_\infty S} \int_{-b/2}^{b/2} \Gamma_0 \sqrt{1 - \left(\frac{2y}{b}\right)^2} dy \quad (5.36)$$

$$= \frac{2}{U_\infty S} \frac{\Gamma_0 b \pi}{4} = \frac{\pi b^2}{S} \underbrace{\frac{\Gamma_0}{2bU_\infty}}_{\epsilon} = \frac{\pi \epsilon b^2}{S} \quad (\text{TEq. (7.12)}) \quad (5.37)$$

Elliptical lift distribution

Aspect ratio:

$$AR = \frac{b}{c} = \frac{b^2}{bc} = \frac{b^2}{S} \quad (5.38)$$

Therefore,

$$\epsilon = \frac{C_L}{\pi AR} \text{ for elliptical loading} \quad (\text{TEq. (7.14)}) \quad (5.39)$$

$$C_{Dv} = \frac{2}{U_\infty S} \int_{-b/2}^{b/2} \Gamma(y) \underbrace{\epsilon(y)}_{\text{const.}} dy = C_L \epsilon \quad (5.40)$$

$$C_{Dv} = \frac{C_L^2}{\pi AR}, \quad AR \rightarrow \infty, \quad C_{Dv} \rightarrow 0 \quad (2-D \text{ wing/airfoil}) \quad (\text{TEq. (7.17)}) \quad (5.41)$$

Generalized lift distribution

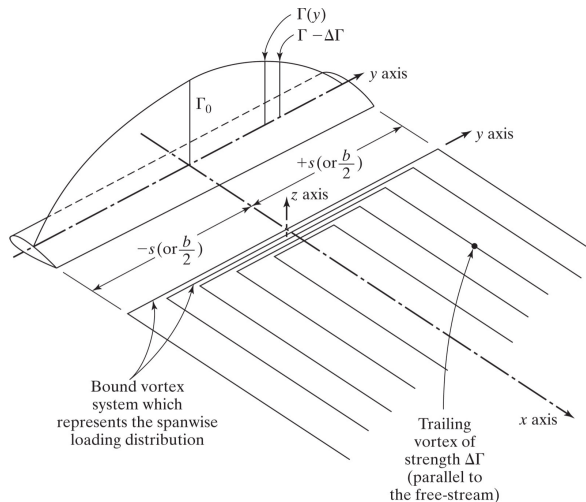


Figure 5.11: PLLT vortex system, Fig. 7.4a from AE-BC

Generalized lift distribution

$$\text{Let } y = -\frac{b}{2} \cos \theta = -s \cos \theta, \quad y_1 = -\frac{b}{2} \cos \theta_1 \quad (5.42)$$

$$\text{and } \Gamma(y) = 2bU_\infty \sum_{n=1}^{\infty} A_n \sin(n\theta) \rightarrow \text{Fourier sine series} \quad (\text{TEq. (7.21)}) \quad (5.43)$$

Note: Elliptic load dist.

$$\Gamma(y) = \Gamma_0 \sqrt{1 - \left(\frac{2y}{b}\right)^2} \quad (5.44)$$

$$\Gamma(\theta) = \Gamma_0 \sqrt{1 - \cos^2 \theta} = \Gamma_0 \sin \theta \rightarrow A_1 = \frac{\Gamma_0}{2bU_\infty}, \rightarrow A_n = 0 \text{ (for } n \neq 1) \quad (5.45)$$

Generalized lift distribution

Substitute the Fourier series into eq. (5.24)

$$\alpha(y_1) = \underbrace{\frac{\Gamma(y_1)}{\pi U_\infty c(y_1)} + \alpha_{ol}(y_1)}_{\alpha_e} + \underbrace{\frac{1}{4\pi U_\infty} \int_{-b/2}^{b/2} \left[\frac{d\Gamma}{dy} \right] \frac{dy}{(y_1 - y)}}_{\epsilon} \quad (5.46)$$

After carrying out the integrals we have

$$\alpha(\theta_1) = \frac{2b}{\pi c(\theta_1)} \sum_{n=1}^{\infty} A_n \sin(n\theta_1) + \alpha_{ZL}(\theta_1) + \sum_{n=1}^{\infty} n A_n \frac{\sin(n\theta_1)}{\sin(\theta_1)} \quad (5.47)$$

Truncate the series to N terms which gives N unknowns.

$$\sum_{n=1}^{\infty} \rightarrow \sum_{n=1}^N$$

Generalized lift distribution

We enforce the equation at N spanwise locations

$$\theta_{1n} = \frac{n\pi}{N+1}, \quad n = 1, 2, \dots, N$$

We avoid $\theta_1 = 0$ and π as $\sin \theta_1 = 0$ there. We then solve the resulting N linear equations for the N unknowns. Once A_n 's are calculated then,

$$C_L = \frac{2}{U_\infty S} \int_{-b/2}^{b/2} \Gamma(y) dy = \frac{2b^2}{S} \sum_{n=1}^N A_n \underbrace{\int_0^\pi \sin(n\theta) \sin \theta d\theta}_{\begin{cases} = 0, & n \neq 1 \\ = \frac{\pi}{2}, & n = 1 \end{cases}} \quad (5.48)$$

$$C_L = A_1 \pi \frac{b^2}{S} = A_1 \pi AR \quad (\text{TEq. (7.27)}) \quad (5.49)$$

Generalized lift distribution

$$C_{Dv} = \frac{2}{U_\infty S} \int_{-b/2}^{b/2} \Gamma(y) \epsilon(y) dy \quad (5.50)$$

$$= 2 \frac{b^2}{S} \int_0^\pi \left(\sum_{n=1}^N A_n \sin(n\theta) \right) \left(\sum_{m=1}^N m A_m \sin(m\theta) \right) d\theta \quad (5.51)$$

as

$$\epsilon(y_1) = - \int_{-b/2}^{b/2} \left(\frac{d\Gamma}{dy_0} \right) \frac{dy}{4\pi U_\infty (y_1 - y)} \quad (5.52)$$

$$C_{Dv} = 2 \frac{b^2}{S} \sum_{n=1}^N \sum_{m=1}^N m A_n A_m \underbrace{\int_0^\pi \sin(n\theta) \sin(m\theta) d\theta}_{\begin{cases} = 0, & n \neq m \\ = \frac{\pi}{2}, & n = m \end{cases}} \quad (5.53)$$

Generalized lift distribution

Therefore,

$$C_{Dv} = \pi \frac{b^2}{S} \sum_{n=1}^N n A_n^2 = \underbrace{\pi A R A_1^2}_{=C_L^2/(\pi A R)} \left[1 + \sum_{n=2}^N n \left(\frac{A_n}{A_1} \right)^2 \right] \quad (5.54)$$

$$= \underbrace{\frac{C_L^2}{\pi A R}}_{C_{Di-ellip}} (1 + \delta) \quad \text{where } \delta = \sum_{n=2}^N n \left(\frac{A_n}{A_1} \right)^2 \geq 0 \quad (5.55)$$

$$\text{Let } e = \frac{1}{1 + \delta}, \text{ then} \quad (5.56)$$

$$C_{Dv} = \frac{C_L^2}{\pi e A R} \quad (\text{TEq. (7.29)}) \quad (5.57)$$

$e \rightarrow$ span efficiency factor, $e = 1$ for elliptical loading and $e < 1$ for any other loading.

Elliptical lift distribution

Hence, the elliptical lift distribution gives the minimum induced drag under the constraint of a fixed wing span b . Note, other constraints such as fixed wing root bending moment, etc. which will lead to other optimal load distributions.

How to achieve elliptical load distribution?

Fundamental Eqn. of Prandtl's lifting line theory:

$$\alpha(y_1) = \frac{\Gamma(y_1)}{\pi U_\infty c(y_1)} + \alpha_{ol}(y_1) + \underbrace{\frac{1}{4\pi U_\infty} \int_{-b/2}^{b/2} \left[\frac{d\Gamma}{dy} \right] \frac{dy}{(y_1 - y)}}_{\epsilon(y_1)=\text{const}} \quad (5.58)$$

Elliptical lift distribution

1. Choose $c(y_1)$ = elliptical **elliptically shaped planform** , $\alpha(y_1)$ = constant, $\alpha_{0l}(y_1)$ = constant.
2. Choose $c(y_1)$ = constant **rectangular planform** , $\alpha(y_1)$ = **elliptical - geometric twist** , $\alpha_{0l}(y_1)$ = constant.
3. Choose $c(y_1)$ = constant **rectangular planform** , $\alpha(y_1)$ = constant, $\alpha_{0l}(y_1)$ = **elliptical - aerodynamic twist** .
4. Combination of above.